

DengueWatch: An Automated Dengue Surveillance and Prevention System



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INTRODUCTION

At present, dengue remains a major public health problem, particularly in tropical and subtropical countries. Dengue is a viral infection primarily transmitted through the bite of infected *Aedes aegypti* mosquitoes. Although many infections are mild or asymptomatic, dengue can progress to severe illness, including severe bleeding, shock, organ impairment, and death. The World Health Organization (WHO, 2025) reported that dengue cases increased dramatically from 505,430 cases in 2000 to 14.6 million cases in 2024, with more than 12,000 dengue-related deaths reported during the same year. This made 2024 the highest annual dengue burden recorded by the WHO.

The increasing spread of dengue is associated with changing mosquito distribution, climate change, increasing temperatures, rainfall, humidity, urbanization, population movement, and limitations in disease surveillance and reporting (WHO, 2025). These factors make dengue prevention increasingly challenging for communities and health systems. At the same time, technological developments provide opportunities to improve disease surveillance and prevention. Artificial intelligence (AI) and machine learning can analyze environmental, climatic, epidemiological, and geographic information to support dengue outbreak forecasting and vector surveillance (Sutanto & Ansharullah, 2025; Geffroy et al., 2026). In this proposal, the author seeks to outline an IT-based solution that can improve dengue monitoring, reporting, and prevention.

PROBLEM DESCRIPTION

Dengue can affect individuals, families, schools, workplaces, and entire communities. One of the major challenges in dengue prevention is the continuous identification and elimination of mosquito breeding sites. *Aedes* mosquitoes commonly reproduce in water-holding containers around homes and communities, making environmental management and community participation important components of dengue prevention (Arfan et al., 2025). Dengue also places pressure on healthcare systems because increasing numbers of cases can require additional healthcare resources. The WHO (2025) reported that dengue cases reached an unprecedented level in 2024, with cases reported in more than 100 countries.

Another problem is the difficulty of obtaining timely and organized information about dengue risks within communities. Delayed reporting can make it difficult for health personnel to

identify areas where mosquito breeding sites or dengue cases are increasing. Sutanto and Ansharullah (2025) explained that technology and AI can support dengue surveillance, outbreak forecasting, and vector-control activities. Furthermore, a review by Geffroy et al. (2026) found that machine learning has been applied to mosquito-borne disease surveillance, including forecasting, risk mapping, and real-time monitoring. With these challenges in mind, there is a need for an accessible technology-based system that allows communities to report potential mosquito breeding sites while providing health personnel with organized information for surveillance and intervention.

II. PROPOSAL SOLUTION

The proposed solution to address dengue is the development of an automated dengue surveillance and prevention system called DengueWatch. This system is designed to help community members and authorized health personnel monitor potential dengue risks through digital reporting, GPS-based mapping, automated notifications, and data analysis. Community members can report possible mosquito breeding sites by submitting the location, photograph, type of breeding site, date, and description. The system can also provide educational materials about dengue transmission, symptoms, warning signs, prevention, and the importance of seeking medical attention. These features support the need for stronger community participation and technology-assisted dengue prevention (Arfan et al., 2025).

Another major feature is a GPS-based dengue risk map that can display verified reports of potential breeding sites and dengue-related concerns. This can help health personnel identify areas where reports are concentrated and prioritize environmental inspection and mosquito-control activities. DengueWatch can also incorporate an AI and machine-learning component that analyzes historical dengue cases, reported breeding sites, rainfall, temperature, geographic location, and other environmental information to identify patterns associated with increased dengue risk. AI would serve as a decision-support tool rather than a method for diagnosing individual patients. Recent research supports the use of AI for dengue outbreak forecasting and public-health interventions (Sutanto & Ansharullah, 2025), while machine learning has also demonstrated potential for disease forecasting, risk mapping, and mosquito-borne disease surveillance (Geffroy et al., 2026). The system will primarily target

community members, students, families, barangay health workers, local health offices, and other public-health personnel.

III. CONCLUSION

Dengue remains a serious public health problem affecting millions of people worldwide. The record number of cases reported in 2024 demonstrates the continuing need for effective disease surveillance, mosquito-control activities, community participation, and early response (WHO, 2025). The problem is further complicated by climate variability, urbanization, changing mosquito distribution, and limitations in traditional surveillance systems. Therefore, addressing dengue requires not only health education but also practical methods for identifying, reporting, and monitoring potential dengue risks within communities.

The proposed DengueWatch automated dengue surveillance and prevention system provides an IT-enabled approach to these challenges by integrating digital reporting, GPS-based mapping, automated notifications, health education, and AI-assisted data analysis. Recent studies support the potential of AI and machine learning for dengue forecasting, risk mapping, and disease surveillance (Sutanto & Ansharullah, 2025; Geffroy et al., 2026). However, technology alone cannot eliminate dengue. Effective prevention still requires community participation, environmental management, mosquito-control activities, and timely healthcare (Arfan et al., 2025). By combining information technology with established dengue-prevention practices, DengueWatch can provide a more organized, responsive, and data-driven approach to dengue prevention and surveillance in communities.

IV. REFERENCES

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